Course Name: Computer Networks

Course Outcome

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

Printed Pages: 04 **University Roll No. .....**

End Term Examination, Even Semester 2022-23
B.Tech (CSE), 2nd Year, IV Semester
BCSC0008 Computer Networks

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Do signal really travel faster in fiber optics? What are some of the uses of fiber optic cabling in the transmission medium and what principle does fiber optic work on?	4	2	U	C
2	Suppose a seven-bit data word 1101110 is to be encoded using even-parity Hamming code. What is the binary value after encoding? Now assume that the 6th bit got corrupted during transmission, find the error correcting code.	4	4	A	P
3	What will be sender and receiver window size in case of Go-Back-N and Selective-Repeat Protocols? Justify these values?	4	3	U	F
4	Deliberate the differences between two congestion control mechanism used in computer networks. "Adaptive retransmission of TCP protocol improves the throughput of the network". Give your views in favour or against of this statement.	4	6	U	C
5	DNS uses UDP instead of TCP. If a DNS packet is lost, there is no automatic recovery. Does this cause a problem, and if so, how is it solved? OR How the anonymous FTP is used to provides a method for the general public to access files on	4	7	A	C

	remote sites? Why should there be limitations on anonymous FTP?				

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	In cases where reliability is not of primary importance, UDP would make a good transport protocol. Give examples of specific cases. The following is the dump of UDP header: 0632000D001CE217 • What is the source port number? • What is destination port number? • What is the length of UDP segment? • What is length of Data?	3	6	A	P
7	Consider the three-way handshaking mechanism followed during TCP connection establishment between host P and host Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a connection request message to Q with a TCP segment having SYN bit = 1, SEQ number = X, and ACK bit=0. Suppose Q accepts the connection request. Show how the information is represented in the TCP segment header that is sent by Q to P? How does TCP' 3-way handshake mechanism address the following problems? (a) Delayed arrival of SYN Packet (b) Delayed arrival of ACK Packet	3	6	An	P
8	The only entries in a certain route table are (128.59.28.0/22, port 0), (128.59.28.0/23, port 1) and (128.59.28.0/24, port 2). These entries indicate CIDR network number, the prefix indication and the corresponding port to which a packet should be forwarded. If a packet arrives with a destination IP address equal to 128.59.29.18, which port will this router forward the packet to?	3	5	An	P

9	An entry in the switching table of a virtual-circuit network is normally created during the setup phase and deleted during the teardown phase. In other words, the entries in this type of network reflect the current connections, the activity in the network. In contrast, the entries in a routing table of a datagram network do not depend on the current connections; they show the configuration of the network and how any packet should be routed to a final destination. The entries may remain the same even if there is no activity in the network. The routing tables, however, are updated if there are changes in the network. Can you explain the reason for these two different characteristics? Can we say that a virtual-circuit is a connection oriented network and a datagram network is a connection Less network because of the above characteristics?	3	5	A	C
10	Suppose a computer sends a packet at the network layer to another computer somewhere in the Internet. The logical destination address of the packet is corrupted. What happens to the packet? How can the source computer be informed of the situation?	3	5	R	C

Section – C

*Attempt **All** Questions*

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. What is the valid allocation of addresses to A and B?	5	5	A	P
12	Illuminate the applications of RSA Algorithm, and perform encryption and decryption using RSA algorithm for p=17, q=11, e=7, M=88.	5	7	A	P

13	Suppose a TCP message that contains 1024 bytes of data and 20 bytes of TCP header is passed to IP for delivery across two networks interconnected by a router (i.e., it travels from the source host to a router to the destination host). The first network has an MTU of 1024 bytes; the second has an MTU of 576 bytes. Each network's MTU gives the size of the largest IP datagram that can be carried in a link-layer frame. Give the sizes and offsets of the sequence of fragments delivered to the network layer at the destination host. Assume all IP headers are 20 bytes. OR Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the first transmission is 2 MSS and the threshold at the start of first transmission is 8 MSS. Assume that a time out occurs during fifth transmission. Find the congestion window size at the end of tenth transmission.	5	6	A	P

Course Name: ENGINEERING MATHEMATICS III

Course Outcome

CO1-Compute the coefficient of correlation and regression between two variables
CO2- Apply Chi-square test
CO3- Use Laplace transform in solving differential equations
CO4-Know the concept of analytic functions and its applications in engineering
CO5-Understand bilinear transformation and its applications
CO6- Use Cauchy's residue theorem in solving real integrals

Printed Pages: 3 **University Roll No. .....**

End Term Examination, Even Semester 2022-23
B. Tech., Year- II, Semester-IV
BMAS 1103, ENGINEERING MATHEMATICS III

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals, (a) Exactly 3 (b) More than 2 (c) None. (d) More than one individual (s) will suffer a bad reaction.	4	1	An	C
2	The small oscillations of a certain system with two degrees of freedom are given by the following differential equations: $D^2x + 3x - 2y = 0$, $D^2x + D^2y - 3x + 5y = 0$, where $D \equiv \frac{d}{dt}$. If $x = 0, y = 0, x = 3, y = 2$ when $t = 0$, find x and y when $t = \frac{1}{2}$ using Laplace Transform. **OR,** Apply Convolution theorem to evaluate: $\int_0^t u^2 \cos(t-u)\, du$	4	3	E	F

3	Evaluate: $L\left\{\int_0^t \frac{e^{-t}\sin t}{t}\,dt\right\}$.	4	3	E	P
4	Show that the function $f(z) = \frac{x\,y^2(x+iy)}{x^2+y^4}, \quad z \neq 0;\ f(0) = 0$ is not analytic at the origin even though Cauchy-Riemann equations are satisfied there.	4	4	A	P
5	Show that the function $u = e^x \cos y$ is harmonic. Also, find its harmonic conjugate. **OR,** Find the bilinear transformation which transforms the points z = 2, 1, 0 into w = 1, 0, i.	4	4	An	F

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Find the analytic function f (z) = u + iv, given that v = e^x (x sin y + y cos y).	3	5	An	F
7	Write a short note on singularity and its types.	3	4	F	C
8	Evaluate $\int_0^{2+i} z^2\,dz$ along the real axis from z = 0 to z = 2 and then along a line parallel to y- axis from z = 2 to z = 2 + i.	3	4	E	P
9	Prove that: $\int_c \frac{z+4}{z^2+2z+5}dz = 0$, if c is the circle $\lvert z+1\rvert = 1$.	3	4	E	C

10	Show that: $\oint_c \frac{dz}{(z^2+4)^2} = \frac{\pi}{16}$, $c:\lvert z-i\rvert=2$	3	4	An	F

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Expand $f(z)=\frac{z}{(z-1)(2-z)}$ in Laurent's series valid in (*i*) $\lvert z-1\rvert>1$ (*ii*) $0<\lvert z-2\rvert<1$.	5	4	C	M
12	Show that the function $f(z)=\frac{1}{z^2-1}$ has simple poles at z = 1 and z = – 1. What will be the residues of f (z) at both these poles? Also, find $\int_c f(z)\,dz$ if c is a circle with centre at origin and radius 1.	5	6	An	P
13	Evaluate the integral $I=\oint_c \frac{e^z}{(z+1)^n}dz,\ c:\lvert z\rvert=2.$ using Cauchy's residue theorem given that n is a constant. **OR,** Evaluate the integral $I=\int_0^{2\pi} \frac{d\theta}{a+b\sin\theta};\ a>\lvert b\rvert$ using calculus of residues.	5	6	A	F

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive).

Course Name: **DATA STRUCTURE & ALGORITHM**

Course Outcome

CO1- Understand the application of data structure and algorithms.
CO2- Implement of Linear Data Structure and its applications.
CO3- Implement of Graph Data Structure and its applications.
CO4- Implement of Trees Structure and its applications.
CO5- Implement of Sorting and Searching Techniques and its applications.

Printed Pages: 3 **University Roll No. .....**

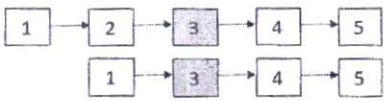
End Term Examination, Odd Semester 2022-23
B. TECH (CS), 2nd Year, IV-SEM
BCSC 0006- DATA STRUCTURE & ALGORITHM

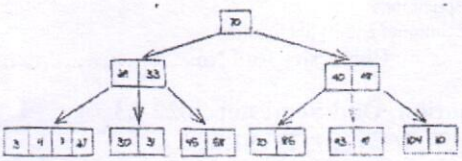
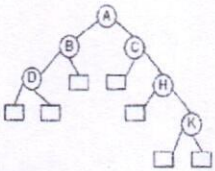
Time: 3 Hours **Maximum Marks: 50**

Instruction for students:
1. Read questions Carefully.
2. Don't Overwrite.
3. Attempt the All parts of Question in one place.
4. Highlight the question number clearly.
5. Commenting the code is optional.

Section – A

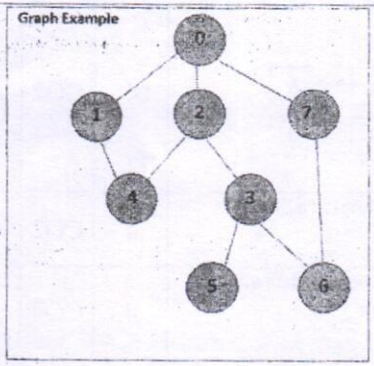
*Attempt **All** Questions* 4 X 5 = 20 Marks

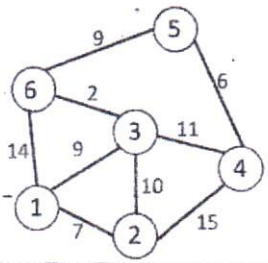
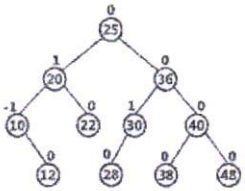
No.	Detail of Question	Marks	CO	BL	KL
1.	Given a Scenario: You are GLA Student Representative. You are tasked to conduct Round Table Conference for the foreign delegate on the Topic of 'International Trade Laws'. You need to ensure the smooth movement of the delegates across the hall before the start and after the end of the Conference. Using the knowledge of Data Structure that you have learned during this Semester. **Identify** the Best way to achieve your task and **Show** the diagrammatic representation for **10 delegates**.	2+2	CO1	U	C
2.	Write a Java Program to delete the previous node of the given node in Single linked List. 	4	CO2	A	P
3.	Write a Java Program to implement FIFO concept in a Single Linked List.	4	CO2	C	M
4.	Obtain the Min Heap and Max Heap of the following data in array: 100, 60, -20, 40, 50, 30, 60, -30, 80	4	CO4	A	P

5.	Identify the **Tree's name** and State the **Rules** while creating such a tree.  Or Identify the **Tree's name** and State the **THREE** applications such a tree. 	4	CO4	An	F

Section – B

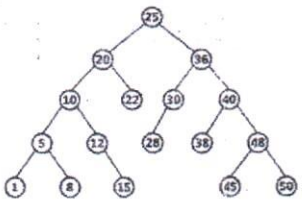
Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6.	Diagrammatically **Sort** the data in given array using the **Divide and Conquer Aigo**: (Represent all the intermediate steps) J, G, K, I, A, Q, N, V, S, W	3	CO5	E	P
7.	State the difference between the two Standard Graph Traversals. **Give the Traversal order** the following graph, using the two Standard Graph Traversal Approach. 	1+2	CO3	R,E	C

8.	Implement the Kruskal Algo, to obtain the MST of following (Show all the intermediate Steps): 	3	CO3	E	P
9.	Provide the PreOrder, InOrder, PostOrder Traversal of the following AVL Tree. 	3	CO4	An	C
10.	Implement the Java program to obtain the height factor of AVL Tree.	3	CO4	U,C	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11.	Quick Sort is one of the fastest sorting Algo given the case of finding the Best-Case Scenario. Implement a Java Program for this Sorting technique to achieve to sort the array data {4, 2, 1, 5, 7, 8, 3, 9, 10} in minimum time.	5	CO5	A	M
12.	State three popular Hashing Techniques. Using Systematic Diagrammatic Representation, Show the Binary Search Operation on the sorted Array to find element 'K'. Array: A, B, C, D, E, F, G, H, I, J, K, L, M	3+2	CO4	R,E	C
13.	Perform the following command on the given BST and show the Tree after every Command execution: a. Search(16) (Write the path while searching for 16) b. Insert(39) c. Delete(40) d. Insert(21) e. Insert(29) f. Insert(16) g. Delete(30) h. Insert(49) i. Delete(25) j. Insert(-50) 	5	CO4	A	P

Course Name: **Introduction to Microprocessor**

Course Outcome

CO1: Demonstrate the Microprocessor internal architecture and its operations.
CO2: Develop programs based on 8085 microprocessor instruction set and addressing mode.
CO3: Develop a program using looping, counting, indexing, counter, and time delays.
CO4: Understand the concept of stack and subroutine for the modular approach.
CO5: Compare accepted standards and guidelines to select microprocessor (8085 & 8086) to meet performance requirements.
CO6: Analyze the concept of interfacing the processor to external device with I/O programming & Interrupt Driven I/O.
CO7: Understand the working of interfacing chips (8237, 8253/54, 8255 & 8259).

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End Term Examination, Even Semester 2022-23
B.Tech (CSE/AIML/DA/IIoT/CCV/CSF), II-Year, IV-Semester
BCSC0007: Introduction to Microprocessors

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	An 8-bit number is stored in memory location 2000H. Count the number of 1's in that number using assembly language programming and save the result in memory location 3000H.	4	3	A	C
2	What is the purpose of providing registers in a microprocessor? Describe the various registers which are usually provided in a microprocessor with the help of a diagram.	4	1	R	F
3	Differentiate between the following instructions: (i) RAL and RLC (ii) CMP and SUB	4	2	U	F
4	Programmable Interrupt Controller can handle various interrupt requests. List the major components used to process these requests and explain their functions.	4	7	U	C
5	Explain different types of addressing modes in 8086 microprocessors with appropriate examples.	4	5/7	An	C

	Or Set up the 8255 (Programmable Peripheral Interface) I/O ports in different modes. Define these modes with suitable examples.				

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Write a BSR control word subroutine to set bits PC_2 and PC_5 and reset them after 15ms. Assume a delay subroutine is available.	3	7	U	P
7	A microprocessor is connected with I/O devices through interfacing. What is the need of interfacing? Differentiate memory mapped and input/output mapped for I/O devices.	3	6	An	C
8	Draw the status and control bit pattern for the flag register of 8086 and explain the significance of each bit.	3	5	An	P
9	8086 operates in two modes. What is the significance of the following signals in the maximum mode of the 8086 microprocessor? (a) NMI (b) TEST' (c) BHE'	3	5	R	F
10	What are the minimum mode and maximum mode in the 8086 microprocessors? Enlist all signals in the proper order.	3	5	U	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Write a program to count from 0 to 9 with one second of delay between each count. At the count of 9, the counter should reset itself to 0 and repeat the counter sequence continuously. Use register-pair HL to set up a delay and display each count at one of the output ports. Assume the clock frequency of the microprocessor is 1 MHz.	5	3	An	P

12	How 20-bit physical memory address is generated in the 8086 microprocessor? Also, briefly describe the memory segmentation with advantages in the 8086 microprocessor.	5	5	An	C
13	8254 operates in different modes. Name them. Also, briefly describe the functions of pins of the 8254 programmable interval timer using an appropriate diagram.	5	7	U	F

Course Name: Discrete Mathematics

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real-world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5: Analyze the validity of an argument using logical notation.
CO6: Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7: Understand the basic principles of sets, set equalities and operations in sets.
CO8: Apply counting principles to determine probabilities.

Printed Pages: **University Roll No. .....**

End Term Examination, Even Semester 2022-23
BTech (CSE), II, IV
Subject Code: BCSC1010, Subject Name-Discrete Mathematics

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:
1. All parts of a question should be answered at one place.

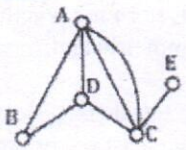
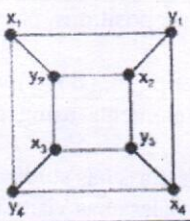
Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Draw the Venn diagram showing the following: (a) $A \cup B \subseteq A \cup C$ but $B \nsubseteq C$ (b) $A \cap B \subseteq A \cap C$ but $B \neq C$	4	CO1	U	F
2	Let R be the binary relation defined as $\{R = (a,b) \in R^2 : a - b \leq 3$ Determine whether R is reflexive, symmetric, antisymmetric and transitive.	4	CO2	R	C
3	Draw the Hasse diagram of the Poset $(D_{24}, \mid)$. Find the greatest lower bound and the least upper bound of the set $\{2, 3, 6\}$ if they exist.	4	CO3	A	C
4	Using laws of algebra of propositions, prove the following $a.\ (p \wedge q) \vee (p \wedge \sim q) \equiv p$ b. $(p \wedge q) \vee p \equiv p$	4	CO3	A	P
5	Express the following statements using predicates and qualifiers: - a. Every student in this class has studied calculus b. Some student in this class has visited Mexico c. Every student in this class has visited either Canada or Mexico.	4	CO5	U	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Find the pcnf and pdnf of $(p \wedge q) \vee (\sim q \wedge r)$ using truth table.	3	CO4	R	F
7	(i) There are 25 telephones in GLA university. Is it possible to connect them with wires so that each telephone is connected with exactly 7 others. Justify. (ii) There are 9-line segments drawn in a plane. Is it possible that each line segment intersects exactly 3 others? Justify.	3	CO3	U	C
8	Let A = {a, b}. Which of the following tables define a semi- group? Which define monoid on A. * | a b a | b a b | a b * | a b a | b b b | a a * | a b a | a b b | b a	3	CO3	A	P
9	Does there exist simple graph with five vertices of the following degree? If so, draw such a graph: (i) 3,3,3,3,2 (ii) 0,1,2,2,3 (iii) 1,1,1,1,1	3	CO4	An	P
10	(i) Does a Hamiltonian path or circuit exist on the graph below?  (ii) Does a graph bipartite 	3	CO4	E	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Which of the following graphs are isomorphic?  G1  G2  G3	5	CO4	An	C
12	Prove the validity of the argument If I work out hard, then I am sore. If I am sore, I take an aspirin. I did not take an aspirin. Therefore, I did not work out hard.	5	CO5	A	C
13	Let G be set of all non-zero real numbers and let $a*b=\frac{ab}{2}$ for a, b $\in G$ show that (G, *) is an abelian group.	5	CO3	U	F

Printed Pages: 18 **University Roll No.....**

End-Term Examination, Even Semester 2022-23

B.Tech. CSE, II & III: Year, IV & VI: Semester

BCSE-0903: Data Science and Analytics

Time: 1.5-hour **Maximum Marks: 50**

SECTION A

Note: Attempt All Questions. **30 x 1 =30 Marks**

1. What is the correct syntax to print number 8 from the array below?

```
arr = np.array([1,2,3,4,5],[3,4,8,2,3])
```

A. print(arr[2,2])
B. print(arr[1,2])
C. print(arr[2,1])
D. None of Above

2.In given code data Frame Df1 has ___rows and ____columns?

```
import pandas as pd
Dict = {"Name": ["Amit" , "Raj" , "Dev"], "RollNo": [1,2,3]}
Df1 = pd.dataFrame(dict)
```

A. 3,3
B. 2,3
C. 3,2
D. None of Above

3. Calculate the interquartile range (IQR) of the following dataset: 1, 2, 3, 4, 5, 6, 7, 8, 9?

A. 1
B. 2
C. 3
D. 4

4 What is the output of the following Python code?

```
x = [1, 2, 3, 4, 5]
y = x [2:4]
print(y)
```

A. [1,2]
B. [2,3]
C. [3,4]
D. [4,5]

5. What is the output of the following Python code?

```
import pandas as pd
data = {"Name": ["John", "Mary", "Peter"], "Age": [25, 30, 35]}
df = pd.DataFrame(data)
print(df.head())
```

A. Name Age
0 John 25
1 Mary 30
2 Peter 35

B. 0 1 2
Name Age
John 25
Mary 30
Peter 35

C. Name Mary Peter
Age 25 30

D. Name Age
John 25
Mary 30
Peter 35

6. In the given chart, parameter surrounded by rectangular box is called?



A. Label
B. X Label
C. Y Label
D. Legend

7. Which of the following is used to change the position of the legend in Matplotlib?

A. loc()
B. position()
C. set_loc()
D. set_position ()

8. How can you select a single column in a DataFrame?

A. df['column_name']

B. df[column_index]
C. df.select_column('column_name')
D. df.column_name

9. Formula for the population variance?
A. (Σ(x-μ)^2)/n
B. Σ(x-μ)^2
C. ΣN/(x-μ)^2
D. (Σ(x-μ) ^2)/N

10. Calculate the standard deviation of the following dataset 2, 4, 6, 8, 10?
A. 2
B. 5
C. 3
D. 3.5

11 What is the output of the following Python code?

```
import numpy as np
arr1 = np.array([1, 2, 3])
arr2 = np.array([4, 5, 6])
arr3 = np.concatenate((arr1, arr2))
print(arr3)
```

A. [1, 2, 3, 4, 5, 6]
B. [1, 4, 2, 5, 3, 6]
C. [4, 5, 6, 1, 2, 3]
D. [1, 2, 3] [4, 5, 6]

12. What is the output of the following code?

```
import numpy as np
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
```

```
c = np.dot(a, b)
print(c)
```

A. 32
B. 16
C. 20
D. 25

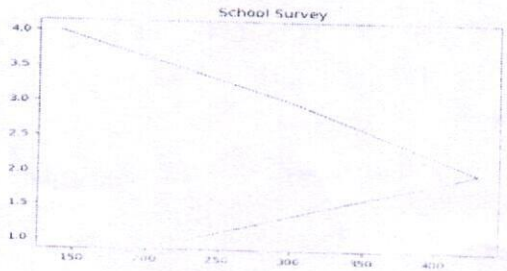
13. How can you concatenate two or more DataFrames?
A. df.concat([df1, df2])
B. df.append([df1, df2])
C. df.merge([df1, df2])
D. df.combine([df1, df2])

14. What is the correct syntax to print the numbers [3,4,5] from the array below?

```
arr = np.array([1,2,3,4,5,6,7])
```

A. print(arr[2:6])
B. print(arr[2:5])
C. print(arr[3:6])
D. print(arr[2:4])

15. Look at the graph and select the appropriate code to obtain this output (Assume that pandas and matplotlib is already imported?



A. zone = [1,2,3,4]
school = [230,340,300,140]
plt.plot(zone, school, School Survey)
plt.show()

B. zone = [1,2,3,4]
school = [230,340,300,140]
plt.plot(school, zone, School Survey)
plt.show()

C. zone = [1,2,3,4]
school = [230,340,300,140]
plt.plot(school, zone)
pt.title('School Survey')
plt.show()

D. zone = [1,2,3,4]
school = [230,340,300,140]
plt.plot(school, zone)
pt.title('School Survey')
plt.show()

16. Raj has written the following code to create a bar chart?

```
import matplotlib .pyplot as plt
Weekdays = ["Sun", "Mon", "Tue"]
Sale = [1234,3312,2451]
plt.plot(weekdays,sale)
```

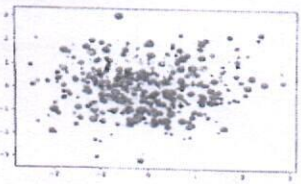
A. He should add plt.show() at the end of the code
B. Instead of plot he should add bar in the plt.plot command
C. He should use numpy array to store weekdays and sales data

D. No suggestion required

17. How can you remove duplicate rows from a DataFrame?

A. df.unique()
B. df.remove_duplicates()
C. df.filter_duplicates()
D. df.drop_duplicates()

18. Identify the following chart type?



A. Bar Chart
B. Histogram
C. Scatter plot
D. Heat Map

19. Calculate the mode of the following dataset: 4, 6, 8, 8, 10?

A. 8
B. 4
C. 6
D. 10

20. What will be the output of the following code?

```
import pandas as pnd

pnd.Series([1,2], index= ['a','b','c'])
```

A. Syntex Error
B. Value Error
C. Index Error
D. None of Above

21 Calculate the Mean of the following dataset: 2, 4, 4, 6, 6, 6, 8, 8, 8, 8, 10?

A. 40.67
B. 88.70
C. 60.90
D. 10.78

22. How many values will be there in array1, if given code is not returning any error?
series4 = pd.Series(array1, index = ["Jan", "Feb", "March" "April"])

A. 1
B. 3
C. 4
D. 2

23. Formula for the population mean?

A. Σx C
B. (Σx)^2
C. (Σx)/N
D. sort[(Σ(x-μ) ^2)/n]

24. Suppose d = {"john":40, "peter":45}. To obtain the number of entries in dictionary which command do we use?

A. d.size()
B. len(d)
C. size(d)

D. d.lean()

25. Rohit has declared a Numpy array: n = np.array([150,160],[80,90])
He wants to make a dataframe with his own column name and index name.
Help him to select correct code?

A. df = pd.DataFrame(n, index = ['A','B'], column = ['C1','C2'])
B. df = pd.DataFrame(index = ['A','B'], column = ['C1','C2'],n)
C. df = pd.DataFrame(n, column = ['C1','C2'] , index = ['A','B'])
D. All are correct

26. Which of the following is used to change the font size of the legend in Matplotlib?

A. Fontsize()
B. font
C. set_fontsize
D. None of the above

27. What is a dimension in Tableau?

A. A measure that is calculated using other measures
B. A categorical or qualitative variable that provides context to a measure
C. A visualization element in Tableau
D. A Tableau feature for combining multiple data sources

28. What is the Output of the following code?

```
import numpy as np
a = np.array([[1, 2], [3, 4]])
print(a)ndim)
```

A. 0

B. 1
C. 2
D. 3

29. What is the range of set of data?

A. The difference between the smallest and largest values in the set
B. The average of the smallest and largest values in the set
C. The difference between the mean and the median of the set
D. The variance of the set

30. Which of the following method is use make the values static of random values array??

A. random. choice ()
B. random.randint()
C. random.seed()
D. random.random()

SECTION B

Note: Attempt All Questions. **40 x 0.5 =20 Marks**

31. How can you get the bottom n rows of a DataFrame?
A. df.tail(n)
B. df.last(n)
C. df.bottom(n)
D. df.rename(columns={'old_name': 'new_name'})

32. What is Numpy?
A. A data analysis library
B. A matrix manipulation library
C. A numerical computation library

D. A data visualization library

33. What is the method to reshape a Numpy array?
    A. ndarray.reshape()
    B. ndarray.resize()
    C. ndarray.shape()
    D. ndarray.modify_shape()

34. What is the method to check for missing values in a pandas DataFrame?
    A. notnull()
    B. isnull()
    C. isna()
    D. notna()

35. Which of the following is the basic data structure in pandas?
    A. Series
    B. Scaler
    C. Vector
    D. Matrix

36. Which of the following method is use to create a scaler value array?
    A. once_like
    B. full
    C. eye
    D. diag –b

37. What is the function to create a Numpy array of ones?
    A. numpy.ones()
    B. numpy.one_array()
    C. numpy.create_ones()
    D. numpy.full_array()

38. Which of the following attribute of DataFrame is used to display all the values from DataFrame?
    A. values
    B. Values
    C. val
    D. Val

39. The data label associated with a particular value of Series is called _____?
    A. Data value
    B. Index
    C. Value
    D. None of Above

40. Which of the following operations can be performed on a Pandas Series?
    A. Filtering
    B. Arithmetic operations
    C. Sorting
    D. All of the above

41. How do you read a CSV file in pandas?
    A. pandas.read_csv()
    B. pandas.load_csv()
    C. pandas.import_csv()
    D. pandas.fetch_csv()

42. What is Pandas?
    A. A programming language
    B. A data analysis library in Python
    C. A machine learning algorithm
    D. A statistical software

43. If a dimension is given as ___ in reshaping operation, the other dimensions are automatically calculated?
    A. Zero
    B. One
    C. Negative One
    D. Infinite

44. Numpy package is cable to do the fast operations on array?
    A. False
    B. True
    C. None of Above
    D. Both A and B

45. Which of the following is the method to change the data type of an array?
    A. dtype
    B. datatype
    C. Astype
    D. astype

46. Which of the following is not an argument of arrange method?
    A. start
    B. stop
    C. end
    D. add

47. Which of the method is used to check the dimension of an array
    A. Shape
    B. dim
    C. shape
    D. Dim

48. What is the method to find the sum of elements in a Numpy array?
    A. array.sum()
    B. numpy.sum()

C. array.add()
D. numpy.total()

49. What is the function to create a Numpy array?
A. numpy.array()
B. numpy.asarray()
C. numpy.create_array()
D. numpy.make_array()

50. What is the function to create a Numpy array of zeros?
A. numpy.zeros_array()
B. numpy.zeros()
C. numpy.empty_array()
D. numpy.create_zeros()

51. What is the method to find the data type of a Numpy array?
A. ndarray.item_type
B. ndarray.type
C. ndarray.data_type
D. ndarray.dtype

52. Which of the following command s used to create a series?
A. createSeries()
B. series()
C. create_series()
D. Series()

53. A _____ is a two-dimensional labelled data structure?
A. DataFrame
B. series
C. List
D. Tuple

54. A_____ is a one-dimensional Array?
A. DataFrame

B. Series
C. Both of Above
D. None of Above

55. Which of the following is not a type of data analytics?
A. Descriptive
B. Subscripted
C. Predictive
D. prescriptive

56. Which of the following command is used to install pandas?
A. pip install pandas
B. install pandas
C. pip pandas
D. None of Above

57. What is the method to fill missing values in a pandas DataFrame or Series?
A. update()
B. replace()
C. fillna()
D. impute()

58. What is the method to find unique values in a pandas DataFrame or Series?
A. unique()
B. nunique()
C. count_unique()
D. get_unique()

59. How can you get the top n rows of a DataFrame?
A. df.top(n)
B. df.first(n)
C. df.head(n)
D. df.select(n)

60. What of the following is measure of dispersion?

A. Mean
B. Median
C. Mode
D. Standard Deviation

61. How can you drop a column in a DataFrame?
A. df.remove('column_name')
B. df.drop(column_index)
C. df.drop('column_name')
D. df.delete(column_index)

62. What is the difference between a population and a sample?

A. A population is a group of individuals, while a sample is a subset of the population.
B. A population is a subset of a sample, while a sample is a group of individuals.
C. A population is a group of individuals, while a sample is another term for the population.
D. A population is a sample of individuals, while a sample is a group of individuals

63. What is the purpose of statistical inference?

A. To make predictions about a population based on a sample
B. To describe a sample using measures of central tendency
C. To estimate the standard deviation of a population
D. To compare two samples and determine if they are different

64. Which of the following is not a typical data quality issue?

A. Completeness
B. Consistency

C. Complexity
D. Accuracy

65. Which of the following is not a technique used for data pre-processing?
A. Sampling
B. Data Modelling
C. Outlier Detection
D. Dimensionality reduction

66. What is Tableau Public?
A. free version of Tableau Desktop for personal use.
B. A cloud-based platform to share and publish Tableau dashboards and visualizations publicly.
C. A mobile application of Tableau.
D. Tableau feature for scheduling data extracts.

67. Which of the following is not a data type in analytics?
A. Structured
B. Semi-Structured
C. Unstructured
D. Random Data

68. Which of the following is not a measure of central tendency?
A. Mean
B. Median
C. Variance
D. Mode

69. Which of the following is not a commonly used programming language for data analytics?
A. Python

B. R
C. JAVA
D. SQL

70. What is the file extension of a Tableau workbook?

A. .tab
B. .tbw
C. .twbx
D. .tide

CS (Hons) IInd

Course Name: Discrete Mathematical Structures

Course Outcome

CO1- Understand the notion of mathematical thinking, mathematical proofs, and algorithmic thinking, and be able to apply them in problem solving.

CO2-Understand the basics of discrete probability and number theory, and be able to apply the methods from these subjects in problem solving.

CO3-Use effectively algebraic techniques to analyze basic discrete structures and algorithms.

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End Term Examination, Even Semester 2022-23
B. Tech. (Hons.), Year II, Semester IV
BCSC0058: DISCRETE MATHEMATICAL STRUCTURES

Time: 3 Hours **Maximum Marks: 45**

All parts of a question should be answered at one place.

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	**a.** Let A, B and C be sets such that $A \subseteq B$, $A \subseteq C$, $B \not\subseteq C$ and $C \not\subseteq B$. Draw the corresponding Venn diagram. **b.** Two finite sets have m and n elements. The total number of subsets of the first set is 56 more the total number of subsets of the second set. Find the values of m and n.	2+2	CO1	U	F
2	Let $A = \{1,2,3,4,5,6\}$. Consider the relation R on A given by iRj iff $(i)\|i-j\| = 2$ $(ii)\|i-j\| < 2$ $(iii)\ \|i-j\| \leq 2$ Test, whether R is (a) reflexive (b) symmetric (c) transitive (d) equivalence? **Or** Let A= {1, 2, 3, ..., 11, 12}, and let R be a relation on A x A defined by $(a, b)\ R\ (c, d)$ if and only if $a+d=b+c$. Prove that R is an equivalence relation	4	CO1	U	C
3	**a.** Given the truth values of p and q as T and those of r and s as F, find the truth values of the following: $(\neg (p \wedge q) \vee \neg r) \vee ((q \leftrightarrow \neg p) \rightarrow (r \vee \neg s))$ **b.** If p denotes "he is rich" and q denoted "He is happy". Write the statement in symbolic form using p and q:If he is rich, then he is unhappy.	2+2	CO2	An	C

4	State the converse, inverse and contrapositive of the following statement: If a triangle is not equilateral then it is not scalene.	4	CO2	A	F
5	Determine whether the set of real number R under the binary operation * defined as a*b=a+b+2 for a,b ∈R is an abelian group or not.	4	CO3	E	P

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Let M(x) be "x is a mammal". Let A(x) be "x is an animal". Let W(x) be "x is warm blooded". Translate into a formula: "Every mammal is warm blooded".	3	CO2	An	C
7	Obtain CNF and DNF of $(p \wedge q) \wedge (q \rightarrow p)$	3	CO2	U	F
8	Check whether the statement $(p \wedge q) \wedge \sim(p \vee q)$ is tautology or contradiction or contingency?	3	CO2	U	F
9	Prove that (G, X) where G={1,-1,i,-i}, i=$\sqrt{-1}$ and 'X' is the operation of ordinary multiplication, is an abelian group.	3	CO3	E	P
10	Draw the Di/graph for the following adjacency matrix: (i) $\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 3 & 2 & 2 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 2 & 2 & 1 & 0 \end{bmatrix}$	3	CO3	R	P

Section – C

Attempt All Questions 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	**a.** Three groups of children contain 3 girls and 1 boy, 2 girls and 2 boys, 1 girl and 3 boys. One child is selected at random from each group. Show that chance that the three selected will consist of 1 girl and 2 boys is $\frac{13}{32}$. **b.** In a family husband tells lies in 30% cases and wife in 35% cases, find the probability that both contradict each other on the same fact?	5	CO2	A	C

12	Find whether the complete graph k_6 and complete bipartite graph $k_{3,3}$ are isomorphic or not. Or Check the validity of the following argument: Either Ram will cook or Hari will practice karate. If Hari practices karate, then Ram studies. Ram does not study. Therefore, Hari will practice karate.	5	CO3	An	P

Course Name: **English for Professional Purposes II (BELH0004)**

Course Outcomes:

CO1-The topics dealing with communication aspects will train the students hands on, by enabling them understand/identify and appreciate the barriers to communication, ways to overcome, process, type of communication etc.

CO2-The students will be able to understand the needs and requirements of the target audience based on their exposure and knowledge of the subject and will be able to communicate accordingly with certain level of accuracy.

CO3-The writing skills of the students will be enhanced to such an extent that they could easily prepare and manage materials and strategies of formal correspondence themselves.

CO4-The students will be equipped with interpersonal skills through listening and speaking activities. The exercises will help in getting rid of stage-fear, nervousness and hesitation during deliveries of official nature.

CO5-The students will be able to identify and rectify grammatical errors in written communication.

Printed Pages:7 **University Roll No. .....**

End Term Examination, Even Semester 2022-23
B. Tech (CS, EC, EE, ME, CE, Biotech), Second Year, IV Semester
English for Professional Purposes II: BELH0004

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What do you understand by 'thought/sense group'? Read the following sentences, and identify (1) the most likely thought/sense group boundaries and (2) the focus word in each thought group. A. If you bring a glass of water, I will give you a chocolate. B. The first item on our agenda is to address parking. C. Joe Biden, the President of the USA, recently started a project to fight poverty. D. Meet my friend, Prof Mishra.	4	4	A	P
2	Write an essay on ***any one*** of the following topics (300 words): A. Impact of increasing inflation on global economy	4	3	A	P

	B. Growing Intolerance in the world C. Atmanirbhar Bharat				
3	**(i) Write the following phonetically transcribed words in English Alphabet:** A. /ˈsæʃeɪ/ B. / kəmˈpjuːtə(r)/ C./ teə(r)/ D. / pəˈteɪtəʊ / E. /ədˈvɑːns/ F. /ˌendʒɪˈnɪə(r)/ G. /əˈsjuːm / H. /ˈtiːtʃə(r)/ **(ii) As you know, while speaking we use syllables. How many syllables do you find in the sentences given below:** A. computer B. number C. count D. following E. corporate F. university G. write H. email	2+2	4	An	P
4	Assuming a job position from your respective field, prepare a tailored resume. **Note**: Don't forget to mention the job position, name of the company, and the key skills required.	4	2	An	P
5	On the basis of the resume you prepared for the answer of Question no. 4 of Section A of this question paper, answer the following interview questions: A. Introduce yourself. B. Why do you think you are the best candidate for this post?	4	4	U	P

Section – B

*Attempt **All** Questions*

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	**It is possible that the following sentences contain error(s). Identify the error(s) wherever they occur in the sentences given below:** (i) The girl's parents (A)/ forbade her to (B)/ not marry the boy. (C)/ No Error (D) (ii) Some people think (A) / acoustic recordings are superior (B)/ than digital ones. (C)/ No Error (D) (iii) The report attacks on (A)/ the idea of exams (B)/for seven and eight-year-olds. (C)/ No Error (D)	3	5	U	C
7	**It is possible that the following sentences contain error(s). Identify the error(s) wherever they occur in**	3	5	U	C

	the sentences given below: (i) Let's face it – (A)/ you're going to be late (B)/ if you go by bus or train. (C)/ No Error (D) (ii) Either you or the teacher (A)/are going to (B)/prepare dinner tonight. (C)/ No Error (D). (iii) It really is one of life's greatest mystery (A)/ why that awful restaurant (B) is busy every night (C)/ No error (D)				
8	**It is possible that the following sentences contain error(s). Identify the error(s) wherever they occur in the sentences given below:** (i) The principal is waiting (A)/ for the students (B)/ since morning. (C)/ No Error (D) (ii) The guests will have eaten (A)/ the dinner by the time (B)/ Ramesh will return home. (C)/ No Error (D)/ (ii) Scarcely had Ramesh sat down to have supper (A)/ then he heard the cries (B)/of the children coming from outside. (C)/ No Error (D)	3	5	U	C
9	**Fill in the blanks with appropriate choices:** (i) It was predicted that a comet would collide …… one of the planets. a) against b) into c) with d) under (ii) We refrained …. talking until we knew that it was safe a) by b) on c) from d) because of (iii) Are the surgeons going to operate …… the badly wounded man? a) For b) with c) at d) on	3	5	U	C
10	**Fill in the blanks with appropriate choices:** (i) His words to the press were deliberately …… - he didn't deny the reports but neither did he confirm them. a) equivocal b) reticent c) infallible d) sustainable (ii) Foreign aid is badly needed to …… the effects of the drought. a) upstart b) worsen c) ameliorate d) simulate (iii) The court's ruling is …… . a) inimitable b) illegible c) amateur d) irrevocable	3	5	R	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.		Marks	CO	BL	KL
11	**The sentences given in each question, when properly sequenced, form a coherent paragraph. Choose the most logical order of sentences to construct a coherent paragraph.** **(i)** A. The implications of retelling of Indian stories, hence, takes on new meaning in a modern India. B. The stories we tell reflect the world around us. C. We cannot help but retell the stories that we value — after all, they are never quite right for us — in our time. D. And even if we manage to get them quite right, they are only right for us — other people living around us will have different reasons for telling similar stories. E. As soon as we capture a story, the world we were trying to capture has changed. A) ECDAB B) BECDA C) BCEDA D) CABDE **(ii)** A. Before plants can take life from atmosphere, nitrogen must undergo transformations similar to ones that food undergoes in our digestive machinery. B. In its aerial form nitrogen is insoluble, unusable and is in need of transformation. C. Lightning starts the series of chemical reactions that need to happen to nitrogen, ultimately helping it nourish our earth. D. Nitrogen — an essential food for plants — is an abundant resource, with about 22 million tons of it floating over each square mile of earth. E. One of the most dramatic examples in nature of ill wind that blows goodness is lightning. A) ECABD B) CDBAE C) ECDBA D) BDECA **(iii)** A. The process of handing down implies not a passive transfer, but some contestation in defining what exactly is to be handed down. B. Wherever Western scholars have worked on the Indian past, the selection is even more apparent and the inventing of a tradition much more recognizable. C. Every generation selects what it requires from the past	5	5	U	C

and makes its innovations, some more than others.
D. It is now a truism to say that traditions are not handed down unchanged, but are invented.
E. Just as life has death as its opposite, so is tradition by default the opposite of innovation.

A) DACBE B) DAEBC C) ACEDB D) EDACB

(iv) A. Scientists have for the first time managed to edit genes in a human embryo to repair a genetic mutation, fueling hopes that such procedures may one day be available outside laboratory conditions.
B. The cardiac disease causes sudden death in otherwise healthy young athletes and affects about one in 500 people overall.
C. Correcting the mutation in the gene would not only ensure that the child is healthy but also prevents transmission of the mutation to future generations.
D. It is caused by a mutation in a particular gene and a child will suffer from the condition even if it inherits only one copy of the mutated gene.
E. In results announced in Nature this week, scientists fixed a mutation that thickens the heart muscle, a condition called hypertrophic cardiomyopathy.

A) AEBCD B) AEBDC C) BDCAE D) CABDE

(v) A. Four days later, Oracle announced its own bid for PeopleSoft, and invited the firm's board to a discussion.
B. Furious that his own plans had been endangered, PeopleSoft's boss, Craig Conway, called Oracle's offer —diabolical, and its boss, Larry Ellison, a —sociopath.
C. In early June, PeopleSoft said that it would buy J.D. Edwards, a smaller rival.
D. Moreover, said Mr. Conway, —he could imagine no price nor combination of price and other conditions to recommend accepting the offer.
E. On June 12th, PeopleSoft turned Oracle down.
A) CABDE B) BDECA C) CABED D) CADBE

12	**Which sentence is the most preferable in respect of grammar/ meaning and is suitable for formal writing:** **(i).** A. It's too bad. Vibhor isn't with us. If he was here, he could translated the letter for us. B. It's too bad. Vibhor isn't with us. If he were here, he could have translated the letter for us. C. It's too bad. Vibhor isn't with us. If he were here, he could translate the letter for us. D. It's too bad. Vibhor isn't with us. If he had here, he could have translated the letter for us. **(ii).** A. The little girl almost drank some of the furniture polish. If she had actually drunk the polish, she could have got sick or even died. B. The little girl almost drank some of the furniture polish. If she had actually drunk the polish, she could have sick or even died. C. The little girl almost drank some of the furniture polish. If she actually drunk the polish, she could have got sick or even died. D. The little girl almost drank some of the furniture polish. If she had actually drank the polish, she could have got sick or even died. **(iii).** A. You would have no trouble at school if you had done your homework. B. You would have had no trouble at school if you had done your homework. C. You would have had no trouble at school if you did your homework. D. You would had no trouble at school if you had done your homework. **(iv).** A. Mira would find the milk if she looked in the fridge. B. Mira would have found the milk if she looked in the fridge. C. Mira would find the milk if she had looked in the fridge. D. Mira would find the milk if she would have looked in the fridge.	5	5	U	C

	(v). A. If you had spoken louder, your classmates would understand you. B. If you spoke louder, your classmates would understand you. C. If you will speak louder, your classmates would understand you. D. If you spoke louder, your classmates would have understood you.				
13	**Read the Cloze passage and fill the blanks with the most appropriate options:** The word democracy is …(i)…. from a conjunction of two Greek words "demos" meaning people and "kasha" which means rule. Thus democracy is when people choose to rule themselves or in the words of Abraham Lincoln, "government by the people, of the people, for the people". Obviously everyone in the country cannot possible take part in the government so the people elect those by whom they want to be governed. What is criticism? Any opinion against the …(ii)….. scheme is criticism. Criticism can be of two types –constructive and destructive. The purpose of ……(iii)… criticism is to improve the present for the good of all. This is a positive approach. Destructive criticism is usually criticism for the sake of criticism. It is a negative approach which is disconcerting at the individual level and destructive in general. Constructive criticism is a must for all democracies. It encourages …(iv)……. participation in the government, which is the hallmark of any democracy. Moreover, it……(v)…. the politicians to accept responsibility for their actions and gauge the public response. It is a system of feedback by which the public pulse can be felt. When these forms are alive and active we may say democracy thrives." **(i).** A. taken B. derived C. found D. inferred **(ii)** A. existent B. living C. existents D. actual **(iii)** A. devastating B. constructive C. destructive D. productive **(iv)** A. peoples B. local C. public D. general **(v)** A. compels B. dissuade C. oblige D. drags	5	5	U	C

Course Name: Database Management System

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using RA.
CO5: Select the information from a database by formulating complex queries in RA.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

Printed Pages: 05 **University Roll No. .....**

End Term Examination, Even Semester 2022-23
B.Tech (CSE , EC(CS)), II Year, IV Semester
BCSC1003 Database Management System

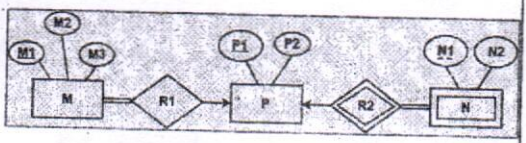
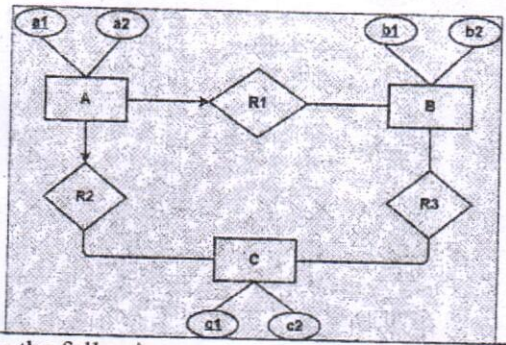
Time: 3 Hours **Maximum Marks: 50**

Instruction for students:
1. Attempt all questions
2. don't switch between the section A, B and Section C, first solve one section then move to second
3. Use only blue pen to answer
4. No B sheet will be given

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the following relational database of a college. **Student**(RollNumber, StudentName, Address) **Teachers**(TeacherID, TeacherName, TeachingSubject) **College**(RollNumber, TeacherID) Write **Relational algebra** expression for the following requests. a. Find the name of Students who live in *Lalitpur*. b. Find the name of teacher who teaches *Database Management System* subject. c. Find the name of teacher who teaches *Computer Organization* subject to *John Smith* student. d. Find subject enrolled by roll no. *786* e. Find name of students taking classes of *Prof. Neeraj Bansal*.	4	CO5	C	P
2	Please create **an Entity Relationship diagram** that captures the information about the College DB. Be certain to indicate identifiers and cardinality constraints. • A college contains many departments	4	CO3	C	P

	• Each department can offer any number of courses • Many instructors can work in a department • An instructor can work only in one department • For each department there is a Head • An instructor can be head of only one department • Each instructor can take any number of courses • A course can be taken by only one instructor • A student can enroll for any number of courses • Each course can have any number of students				
3	Convert into **minimum** number of relations required for the following ER diagram in the relational model a.  b. 	4	CO2	AN	P
4	Answer the following: a. Which normal form is considered adequate for normal relational database design? b. Consider a schema R(A, B, C, D) and functional dependencies A→B and C→D. Test and tell whether the decomposition of R into R1 (A, B) and R2(C, D) is dependency preserving or not?	4	CO6	AN	C

	c. Is this decomposition in part (b) is lossy? (with reason)				
5	Write short note on the following: a. MVD and 4NF b. JD and 5NF	4	CO6	R	F

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider a schema ***R(A,B,C,D)*** and functional dependencies ***A->B and C->D*** which of the following decomposition are dependency preserving: a. ***R1(AC) and R2(BD)*** b. ***R1(AB) and R2(CD)*** c. ***R1(AB), R2(BC) and R3(AD)*** d. ***R1(BC), and R2(AD)*** e. ***R1(AC), R2(BC) and R3(CD)*** f. ***R1(CD) and R2(AC)***	3	CO6	F	C
2	Discuss the following log based recovery: a. Deferred b. Immediate c. Checkpoint	3	CO8	U	C
3	Consider schedules S1, S2, and S3 below. Determine whether each schedule is *strict*, *cascadeless*, *recoverable*, or *nonrecoverable*. (Determine the strictest recoverability condition that each schedule satisfies.) **S1**: $r_1(X); r_2(Z); r_1(Z); r_3(X); r_3(Y); w_1(X); c_1; w_3(Y); c_3; r_2(Y); w_2(Z); w_2(Y); c_2;$ **S2**: $r_1(X); r_2(Z); r_1(Z); r_3(X); r_3(Y); w_1(X); w_3(Y); r_2(Y); w_2(Z); w_2(Y); c_1; c_2; c_3;$ **S3**: $r_1(X); r_2(Z); r_3(X); r_1(Z); r_2(Y); r_3(Y); w_1(X); c_1; w_2(Z); w_3(Y); w_2(Y); c_3; c_2;$	3	CO7	U	C
4	Suppose we have a relation $R(X,Y,Z,W)$ with the FD's: $X \rightarrow Y, Y \rightarrow Z, Z \rightarrow W$ Which one of the following decompositions is not lossless? a. $R_1(X,Y), R_2(Y,Z), R_3(Z,W)$ b. $R_1(X,Y), R_2(X,Z), R_3(X,W)$	3	CO6	AN	P

	c. $R_1(X,W), R_2(Y,W), R_3(Z,W)$				
5	Suppose that transaction T_5, T_{10}, T_{15} have time-stamps 5, 10 and 15 respectively. i. What will happen to T_5 requesting for data item holds by T_{10} in *wait-die* scheme? ii. What will happen to T_{15} requesting for data item holds by T_{10} in *wait-die* scheme? iii. What will happen to T_5 requesting for data item holds by T_{10} in *wound-wait* scheme? iv. What will happen to T_{15} requesting for data item holds by T_{10} in *wound-wait* scheme?	3	CO5	A	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Answer the following: a. Discuss Properties of the transaction b. State diagram of the transaction	5	CO7	U	C
2	Answer the following: a. Discuss two-phase locking protocol. What is the limitation of 2PL? How strict 2PL helpful in overcoming those limitations? What is a locking point? b. What is Distributed database? Discuss fragments and its type with help of example?	5	CO8	U	P
3	Which of the following schedules is conflict serializable? Explain why or why not the schedule is serializable. For each serializable schedule, determine the equivalent serial schedule. i. $r_1(X); r_3(X); w_1(X); r_2(X); w_3(X);$ ii. $r_1(X); r_3(X); w_3(X); w_1(X); r_2(X);$ iii. $r_3(X); r_2(X); w_3(X); r_1(X); w_1(X);$ iv. $r_3(X); r_2(X); r_1(X); w_3(X); w_1(X);$ v. $r_3(X); r_2(X); r_1(X); w_2(X); w_3(X);$	5	CO7	E	M

Course Name:

Course Outcome

CO1: Understand the classification of operating system environment.
CO2: Understand the basic of process management.
CO3: Apply the concept of CPU process scheduling for the given scenarios.
CO4: Illustrate the process synchronization and concurrency process in operating system.
CO5: Analyze the occurrence of deadlock in operating system.
CO6: Describe and analyze the memory management and its allocation policies.
CO7: Understand the concepts of disk scheduling

Printed Pages:3 **University Roll No. .....**

End Term Examination, Even Semester 2022-23

B. Tech. - CS, CCV, DA, CSF, AIML, IIOT, II, IV

Subject Code & Subject Name- BCSC 0004 & Operating Systems

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Make a suitable illustration related to the process state transition diagram for dealing with a specific scenario, where each process is in one of the five states. The operating system uses only non-preemptive scheduling algorithm.	4	2	R	P
2	Consider a system (that follows pre-emption) consists of three processes namely, Process1, Process2, and Process3, where each process has N number of instances, and instance of all the processes arrives at the 0th millisecond. The arrival of the next instances is periodic, with time interval of 3 milliseconds for Process], 7 milliseconds for Process2 and 20 milliseconds for Process3. Every instance of Process], Process2, and Process3 needs CPU burst time of 1, 2, and 4 milliseconds, respectively. The priority of each process is inverse of its burst time. Processes get the CPU in the order of their priority (highest one gets the CPU first). With a suitable GANTT chart, explain the scenario and Compute the following (taking the system as uniprocessor): **i.** Time where 1st instance of the Process3 will get pre-empted. **ii.** Time where 1st instance of the Process3 will complete its execution.	4	3	A	C

3	Give the principles, which should be followed by any solution designed to achieve mutual exclusion in critical section problem. Also, discuss how well these principles are followed in Peterson's solution.	4	4	A	C
4	Given 3 processes A,B and C, three resources x,y and z and following events, i) A requests x ii) A requests y iii) B requests y iv) B requests z v) C requests z vi) C requests x vii) C requests y Assume that requested resources should always be allocated to the request process if it is available. Draw the resource allocation graph for the sequences. And also mention whether it is a deadlock? If it is, how to recover the deadlock.	4	5	E	P
5	Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is _____ megabytes. **Or** Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, find out the effective memory access time (in milliseconds). Also discuss about TLB.	4	6 6	An E	C C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	How do caches help improve performance? Why do systems not use more or larger caches if they are so useful?	3	6	An	C
7	Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.	3	6	U	P
8	The seek time will be reduced in which disc scheduling technique, Explain the difference between internal and external fragmentation.	3	7	An	C
9	Discuss the sequential, indexed and random file access schemes. Which allocation scheme will minimize the amount of space will minimize the amount of space required in directory structure and why?	3	6	R	F
10	A disk is with 1000 cylinders, numbers 0 to 999; compute the no. of tracks the disk arm must be move to satisfy the entire request in the disk queue. Assume the last request serviced was at track 345 and head is moving towards 0.	3	7	C	PC

	The queue in FIFO order contains request for following tracks: 123, 874, 692, 475, 105, 376 Perform the computation for the following scheduling scheme. A. SCAN B. C-LOOK				

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No	Detail of Question	Marks	CO	BL	KL
11	Describe the Banker's algorithm for safe allocation. Consider a system with three processes and three recourse types and at time T0 the following snapshot of the system has been taken: i. Is the current allocation safe state? ii. Would the following request be granted in the current sate? A. Process P2 request (1,0) B. Process P1 request (1,0)	5	5	E	PC
12	If FIFO page replacement is used with four page frames and eight pages, how many page faults will occur with the reference string 0 1 7 2 3 2 7 1 0 3 if the four frames are initially empty? Repeat this problem for LRU.	5	5	E	C
13	Explain Shortest-seek-time-first algorithm (SSTF) disk scheduling algorithm with any example. Why SSTF scheduling tends to favor middle cylinder over the inner most and outermost cylinder?	5	7	R	F

Process	*Allocated*			*Maximum*			*Available*		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P1	2	2	3	3	6	8	7	7	10
P2	2	0	3	4	3	3			
P3	1	2	4	3	4	4			

Course Name: Cloud and Business Process Management

Course Outcome

CO1-Understand basics of cloud Storage systems.
CO2-Explian the technologies and approaches for the business-related issues.
CO3-Understand the operation view and services catalog of cloud management.
CO4-Understand the concepts of VPM cloud computing.
CO5-Design process interaction interfaces for business users.

Printed Pages:03 University Roll No. .....

End Term Examination, Even Semester 2022-23
B.Tech. (CCV), Year-II, Semester-IV
BCSE 0508 Cloud and Business Process Management

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: Attempt all questions of a section at one place.

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	An organization wants to use BPM. How the business process of the organization would be enhanced by BPM.	4	2	An	C
2	In a cloud environment, the user has a variety of choices for paying the bill. The service provider also provides a variety of metering and billing options. Give a thorough breakdown of the various metering & billing options that a user has and service provider provides.	4	1	U	F
3	For a business, a systematic process flow is needed. How can the BPM life cycle provide better process flow? Give a detailed description of the evaluation of the BPM lifecycle phase.	4	4	U	F
4	An organization plans to run all its services on the cloud platform and wants to adopt a common reference architecture for better BPM. Provide a solution for the organization so that it's all services are easily run on the cloud.	4	5	R	F
5	For better time and effort management, a corporation wishes to leverage an existing business system with as few changes as possible in the software design process. Provide a solution for the above scenario. Or	4	3	A	P

	Design the process that must be performed to apply the docker commit operation to ten files inside a directory and upload them to the docker hub with reference to the business process automation. Include a description with the appropriate steps.	4	5	C	D

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Distinguish between the terms "Coach" and "Coach View "in the view of IBM-BPM also Summaries Heritage Coaches in a brief remark.	3	1	An, R	C
7	A company has chosen to make its cloud solution resilient. Recommend a solution a company may archive resilience for their cloud product. Also, describe the support system's capabilities.	3	3	E, R	M
8	A company wants to choose a integrated cloud service solution. Recommend a way through which "BPM-enabled process automation" will be beneficial to the organization? Also comment on the degree of automation and degree of repetition.	3	2	R	F
9	Over a period, you found that your system performance is deteriorating. Provide a solution that may speed up the processing on your PC. Does the performance of systems depend on patches? Defend your response. Name some tools for patch management.	3	2	E	D
10	A company wishes to fully use BPM for its market advantages. Can sustainability offer a more viable option for implementing market changes? Justify your response.	3	1	U	F

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	In a cloud environment, aim for solutions that will solve issues more quickly. Provide a solution that can solve issues more quickly by using the Aneka container. List the attributes of Aneka that enable resource scalability.	5	4	A, R	C

12	Compare and contrast several self-service provisioning strategies. Detail the Inter-Cloud Operations framework in depth.	5	1	An, R	M
13	An organization is looking to hide the details at different hierarchy levels. Provide the different ways through which the organization achieves abstraction. You can use a diagrammatic representation to support your answer.	5	5	An, R	C

Course Name: Applied Statistical Analysis

Course Outcome

CO1- Understand the different variables measurement and scalling.
CO2- Analyze the different central tendency operations.
CO3- Demonstrate the different ways to represent the data.
CO4- Analyze the hypothesis testing using different estimation and distribution methods.
CO5- Analyze the correlation and regression for the real world problem data.
CO6- Analyze the data using the advanced techniques like, Chi-squared goodness-of-fit test, Chi- square test of independence, the sign test and Mann-Whitney test.

Printed Pages: 4 **University Roll No. .....**

End Term Examination, Even Semester 2022-23
B.Tech (CSE (DA), II[nd] Year, IV Semester
Subject Code & Subject Name- BCSE0552 & Applied Statistical Analysis

Time: 3 Hours **Maximum Marks: 50**

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	The mean annual salary of employees of a company is Rs 30000. The mean annual salaries of male and female employees are Rs 35000 and Rs 23000, respectively. Find out the percentage of male and female employees working in the company.	4	2	U	C
2	The arithmetic mean and standard deviation of a series of 20 items were computed as 20 cm and 5 cm, respectively. While calculating these, an item 13 cm was misread as 30. Find the correct mean and standard deviation.	4	3		C
3	In a frequency distribution, the coefficient of skewness based on quartiles is 0.5. If the sum of the upper and the lower quartiles is 28 and the median is 11, find the values of the lower and upper quartiles.	4	2		P
4	In a certain factory manufacturing razor blades, there is a small chance 1/50 for any blade to be defective. The blades are placed in packets, each containing 10 blades. Using the Poisson distribution, calculate the approximate number of packets containing not more than 2 defective blades in a consignment of 10,000 packets.	4	4		P
5	A manufacturer claimed that at least 95% of the equipment, which he supplied to a factory	4	6		C

	conformed to specifications. An examination of a sample of 200 pieces of equipment revealed that 18 were faulty. Test his claim at a significant level of (a) 0.01, (b) 0.05. (Required data: -2.33,-1.64)				

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	The divisional manager of a retail chain believes the average number of customers entering each of the five stores in his division weekly is the same. In a given week, the manager reports the following number of customers in the stores as: 3,000, 2,960, 3,100, 2,780 and 3,160 Test the divisional manager's belief at 5 per cent level of significance.(Required value:9.488)	3	6	U	C
7	Analyze the below data and tell whether you can conclude that smoking causes cancer or not?	3	6	A	P
8	Find the coefficient of correlation between X and Y, given that, $10Y = 8X + 66$ and $40X = 18Y + 214$ are the regression of Y on X and X on Y, respectively.	3	5	E	P
9	Define F-test? Discuss its application in testing whether the two variances are homogeneous.	3	4	R	F
10	A professor wants to know if her introductory statistics class has a good grasp of basic math. Six students are chosen at random form the call a given a math proficiency test. The professor wants the class to be able to score above 70 on the test. The six students get the following scores: 62, 92, 75, 68, 83, 95. Can the professor have 90% confidence that the mean score for the class on the test would be above 70.(Required value: 2.1328)	3	4	An	C

Category	Diagnosed as Cancer	Without Cancer	Total
Smokers	400	300	700
Non-Smokers	300	500	800
Total	700	800	1500

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

<table>
<tr><th>No.</th><th>Detail of Question</th><th>Marks</th><th>CO</th><th>BL</th><th>KL</th></tr>
<tr><td>11</td><td>The personnel manager of a firm is recruiting its graduate trainees from colleges affiliated to two universities. The performance index (0 - 10 scale) of the trainees from each university follows normal distribution. The variance of performance index of trainees from University I is 9 and that of trainees from University II is 4. The manager feels that the mean performance index of trainees from University I is less than that of trainees from University II. To test his intuition, he has selected a sample of 49 trainees from University I, and their mean performance index is found to be 7. Similarly, he has selected a sample of 64 trainees from University II, and their mean performance index is found to be 6. Test the intuition of the manager at a significant level of 0.05.(Required value : 1.64)</td><td>5</td><td>4</td><td>A</td><td>C</td></tr>
<tr><td>12</td><td>A strength test carried on sample of two yarns spun of the same count gives the following results:
<table>
<tr><th></th><th>Sample Size</th><th>Sample Mean</th><th>Sample Variance</th></tr>
<tr><td>Yarn A</td><td>4</td><td>52</td><td>42</td></tr>
<tr><td>Yarn B</td><td>9</td><td>42</td><td>56</td></tr>
</table>
The strengths are expressed in pounds. Is the difference in mean strengths significant of the sources from which the samples are drawn at 1 per cent level of significance?(Required value: 2.718)</td><td>5</td><td>6</td><td>E</td><td>P</td></tr>
<tr><td>13</td><td>Consider the following ANOVA table, based on information obtained for three randomly selected samples from three independent populations, which are normally distributed with equal variances.</td><td>5</td><td>5</td><td>E</td><td>C</td></tr>
</table>

Source of Variation	Sum of Squares SS	Degrees of Freedom df	Mean Squares MS	Value of the Test Statistic
Between samples			20	
Within samples	60			F=
Total		14		

a) Complete the ANOVA table by filling in missing values.

b) Test the null hypothesis that the means of the three populations are all equal, using 0.01 level of significance.

Course Name: BCSE0602: IT Network Security

Course Outcome

CO1: Understand the fundamentals of network security.

CO2: Demonstrate the principle of network security controls, protocols and devices.

CO3: Design the network security policy for securing any network.

CO4: Understand the concepts related to physical and host security, and secure IDS configuration & Management.

CO5: Conceptualize the concepts of providing network security by using VPN and wireless technologies.

CO6: Explain the concepts related to network traffic monitoring and analysis.

CO7: Conceptualize the risk management framework and concepts of vulnerability management.

CO8: Explain the concepts related to data backup, recovery tools and incident response management.

Printed Pages:3 **University Roll No. .....**

End Term Examination, Even Semester 2022-23
B. TECH. (CSF), Year: II, Semester: IV
BCSE0602: IT NETWORK SECURITY

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Details of Question	Marks	CO	BL	KL
1	Design the Access control, Vulnerability management, e-mail and Data Backup policy for any organization.	4	3	An	P
2	Show the complete process of Digital Signature for message authentication. Which type of encryption technique is used in Digital Signature? Which is the best way of distributing public keys over the network.	4	2	U	C
3	Consider a situation where you were given the task of designing physical security for any organization. Give the various points that you will consider for providing the physical security in such a scenario.	4	4	An	P

4	Define Risk. Prepare a risk matrix that shows the outcome related to the probability of occurrence and consequence.	4	7	U	C
5	On which of the layer of TCP/IP VPN works. Why VPN concentrator is needed for VPN connections. Give all the important functions of VPN concentrator. or How one can differentiate normal and attack traffic signature? Highlight the working of SNMP monitoring using appropriate representation.	4	5	U	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Details of Question	Marks	CO	BL	KL
6	An organization is managing large amount of data on day to day basis. As an expert one need to provide backup solution to such an organization. Identify three different backup types with their advantage and disadvantage for the given situation.	3	8	An	P
7	Any organization wants to provide the Wi-Fi facility to its entire building. Briefly explain the architecture which can be used in such a scenario with appropriate figure. Also, mention which type of antenna should be provided to the access point in such a scenario.	3	5	An	C
8	Discuss the three different approaches which can be used to analyze network traffic to identify possible network attack.	3	6	U	P
9	Discuss the important steps of Vulnerability management for any organization.	3	7	U	C
10	We have to establish VPN for connecting two remote offices of any organization. Which type of VPN is best suited in this case. Also, give the major steps for configuring VPN in such a scenario.	3	5	An	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Details of Question	Marks	CO	BL	KL
11	Compare five different RAID levels by focusing on its advantage and disadvantage.	5	8	U	C
12	Suppose you are leading an Incident Response Team of any organization. One day you realized that some attack took place in your organization. Enlist the important steps related to Incident Response that you will follow in such a situation.	5	8	An	P
13	Give the five most important points that one must consider while dealing with Wireless Network Security.	5	5	U	C

Course Outcome

CO1- Understand the way in which intelligent system is methodologically distinctive while at the same time is an interdisciplinary field where established fields of research – including Psychology, Computer Science, Linguistics, Neuroscience, and Philosophy-contribute to the overall approach.

CO2 – Develop skills in analysing, interpreting, and assessing the empirical data and research techniques that contribute to the intelligent system.

CO3 – Understand central, modelling technique in intelligent, including traditional computational approaches, neural network/deep learning approaches, and dynamical approaches.

CO4 – Learn to communicate technical work in particular research areas in writing and in oral presentation to an interdisciplinary audience.

CO5 – Able to use critical thinking to evaluate and interpret evidence, and to apply psychological concepts, theories, and research findings to individuals, social and cultural issues.

Printed Pages: 03 **University Roll No** .....

End-Term Examination Even Semester 2022-23

B.Tech CSE- AIML, II Year, IV Semester

BCSE 0702 Algorithm for Intelligent system & Robotics

Time: 3 Hours **Maximum Marks: 50**

Section – A

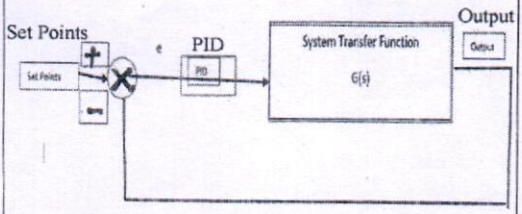
Attempt All Questions 4 X 5 = 20 Marks

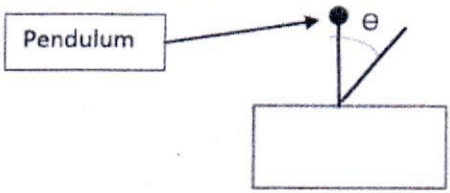
No	Detail of Question	Marks	CO	BL	KL
1	How synaptic plasticity plays a significant role in Human Intelligence.	4	1	U	C
2	Explain with numerical example the difference between monotonic heuristic and consistent heuristic.	4	2	A	P
3	State Bays rule and with respect to some numerical example illustrate the terms posterior probability, likelihood data and prior probability	4	3	A	P
4	Illustrate the rules of data augmentation in machine learning problem.	4	4	U	D

5	In image based application illustrate Horizontal and Vertical shift augmentation. Or In random zoom augmentation how we can zoom the image either by adding new pixel values around the image or by interpolating pixel values?	4	5	E	DI

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider Proportional Integral Derivative (PID) controller which is applied to the following system;  How the error 'e' will be modified? (From initial time instant t = 0 to some finite time t = 10) by the PID controller.	3	2	A	P
7	Given a fourth order differential equation, how to write down four first order differential equations to express it into state space form. $\dot{X} = A_{n \times n} X$ Where $\dot{X} = \begin{vmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{vmatrix}$ Where '–' indicates derivatives.	3	3	U	D

8	Consider the governing equation of an inverted pendulum,  $m \ddot{x} \cos\theta + m l \ddot{y_2} = m g \sin\theta$ Rearrange the equation to obtain the derivative of θ and $\bar{\theta}$ for Rungee – Kutta update. Where '-'indicates derivative.	3	1	U	C
9	How to use Gekko to solve an optimal control problem.	3	2	U	D
10	State the operation of each layer of a Convolution Neural Net (CNN) where input to CNN is an (n X n) image.	3	2	A	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Configure image data for augmentation using Keras so that there is no miss classification.	5	1	E	DI
12	Illustrate with example the geometric transformation of augmenting image data.	5	3	E	C
13	How the data augmentation approach reduces overfitting for machine learning problem.	5	3	U	P

Format of Question Paper for End-Term Examination, B. Tech. (Hons.) /BCA (Hons.)

Course Name:

Course Outcome

- CO1: Understanding of complexity representation in terms of Big Oh, Theta and Omega notations.
- CO2: Derive and solve recurrences describing the performance of divide-and-conquer algorithms (quick sort and merge sort).
- CO3: Compare and analyze different data structures (RB Tree, B Tree, Binomial Heaps, Fibonacci Heaps).
- CO4: Understand the major graph algorithms (DFS, BFS, Dijkstra's Bellman Ford) and their analyses.
- CO5: Understand the greedy paradigm and able to analyze when an algorithmic design situation calls for it. Synthesize greedy algorithms (Optimal Reliability Allocation, Minimum Spanning Trees, factorial Knapsack) and analyze them.
- CO6: Synthesize dynamic-programming algorithms (0/1 knapsack problem, Resource allocation problem, Warshal's and Floyd's algorithms) and analyze them.
- CO7: Understand the backtracking paradigm and able to analysis when an algorithmic design situation calls for it. Synthesize backtracking algorithms (N Queen Problem, TSP Problem, sum of subsets problem, Graph Coloring) and analyze them.
- CO8: Understand the branch and bound paradigm and able to analysis when an algorithmic design situation calls for it. Synthesize branch and bound algorithms (N Queen Problem, TSP Problem, Hamiltonian Cycles, Graph Coloring) and analyze them.

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End Term Examination, Even Semester 2022-23
B.Tech. CS (Hons.), Year II, Semester IV
BCSC: 0059 ALGORITHMS DESIGN & ANALYSIS

Time: 3 Hours **Maximum Marks: 45**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Is the space consumed by the linear search(recursive) and linear search(iterative) the same? Justify? Also prove that $O(n+1)^2 = n^2$	4	1	R	F
2	Lemma: A red-black tree with n internal nodes has a height at most $2 \lg(n + 1)$.	4	3	U	F
3	Show that the running time of QUICKSORT is (n^2) when array A contains distinct elements and is sorted in decreasing order, explain using a suitable example, and analyze the complexity.	4	2	A	C
4	Develop a recursive solution for the matrix Chain Multiplication Problem, Find M [1,4] for the following instances <5,4,6,2,7>.	4	6	An	F
5	How can N queens be placed on an NxN chessboard so that no two of them attack each other? Design a possible solution to the N-queen's problem for N = 5.	4	7	C	P

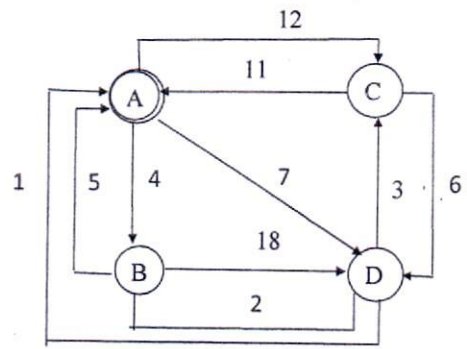
Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Design an algorithm for the scenario where a graph and a source vertex "S" is given in the graph, and find the shortest paths from "S" to all vertices in the given graph. The graph may contain negative weight edges.	3	4	C	C
7	Design an algorithm using the backtracking approach for the scenario where you have given a set of positive integers, and find the combination of numbers that sum to the given value M. Explain the solution using a Tree.	3	7	An	P
8	Given the weights and profits of N items, in the form of {profit, weight} put these items in a knapsack of capacity W to get the maximum total profit in the knapsack. In Fractional Knapsack, we can break items for maximizing the total value of the knapsack. W=50, . Input: arr[] = {{60, 10}, {100, 20}, {120, 30}}	3	5	A	F
9	Suppose that all edge weights in a graph are integers in the range from 1 to |V|. How fast can you make Prim's algorithm run? What if the edge weights are integers in the range from 1 to W for some constant W?	3	5	A	C
10	Given two strings, **S1** and **S2**, the task is to find the length of the longest common subsequence present in both of the strings, using a suitable algorithm. **Input:** S1 = "AGGTAB", S2 = "GXTXAYB"	3	6	U	C

Section – C

*Attempt **All** Questions* 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Design an algorithm and solve using a suitable example for the following scenario The optimal substructure of the problem is given as: $D^k[i,j] = w[i,j]$, if $k=0$ Otherwise $D^k[i,j] = \text{Minimum } \{ D^{k-1}[i,j], D^{k-1}[i,k] + D^{k-1}[k,j] \}$	5	6	C	M
12	Given a set of cities and the distance between every pair of cities, the problem is to find the shortest possible route that visits every city exactly once and returns back to the starting point. Drive a solution using a Branch and Bound approach for the following graph.  	5	8	An	E

Course Name: Machine Learning Using Python

Course Outcome

CO1-Apply the basic concepts of machine learning.
CO2-Analyze the concepts of regression and classification methods.
CO3-Design unsupervised learning-based solutions.
CO4-Analyze the concepts of artificial neural network methods.
CO5-Identify the ways of feature extraction, reduction, and selection.
CO6-Design the applications of machine learning algorithms.

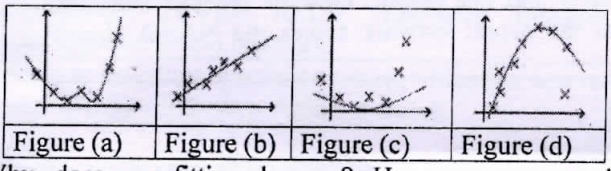
Printed Pages: **University Roll No. .....**

End Term Examination, Even Semester 2022-23
B. Tech. (Hons)- CS, Year: II, Semester: IV
BCSE 0450, Machine Learning Using Python

Time: 3 Hours **Maximum Marks: 45**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

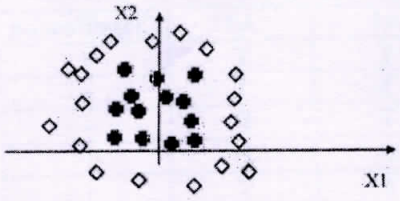
No.	Detail of Question	Marks	CO	BL	KL
1	Why the principle of the gradient descent algorithm used in ML. Accompany your explanation with a diagram. Explain the use of all the terms and constants that you introduce and comment on the range of values that they can take.	4	1	An	C
2	In which one of the following figures do you think the hypothesis has overfit the training set?  Figure (a) Figure (b) Figure (c) Figure (d) Why does overfitting happen? How can you avoid overfitting? **Or** Consider the following set of training examples:	4	2	An	C

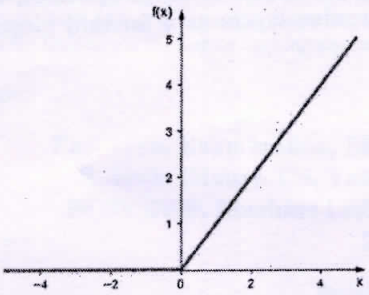
Instance	A1	A2	Classification
1	T	T	+
2	T	T	+
3	T	F	-
4	F	F	+
5	F	T	-
6	F	T	-

	What is the information gain of A2 relative to these training examples? Provide the equation for calculating the information gain as well as the intermediate results.				
3	During a research work, you found 7 observations as described with the data points below. You want to create 3 clusters from these observations using K-means algorithms. After first iteration, the clusters C1, C2, C3 has the following observations: C1: {(2,2), (4,4), (6,6)} C2: {(0,4), (4,0)} C3: {(5,5), (9,9)} If you want to run a second iteration, then what will be the cluster centroids?	4	3	A	P
4	Given an image in the form of numerical matrix, perform average pooling with respect to the following dimensions. Kernel size = 2, Stride = 2, and Padding = None [[2.3, 9.6, 4.3, -7.8], [9.0; 1.1, 3.7, 11.4], [12.5, -2.5, 2.7, 0], [3.6, 3.8, -2.3, 9.8]]	4	4	A	P
5	Given a shallow neural network with two inputs, no hidden layer and one output. Find the weights and bias such that the neural network solves the NAND circuit logic.	4	4	An	C

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Suppose that we want to build a neural network that classifies two-dimensional data (i.e., X = [x1, x2]) into two classes: diamonds and crosses. We have a set of training data that is plotted as follows.  	3	4	An	C

	Draw a network that can solve this classification problem. Justify your choice of the number of nodes and the architecture. Draw the decision boundary that your network can find on the diagram.				
7	 Identify the function and transform the following data according to the function: {10, -25, -18, 0,100, 45, 5.6, -6.7}	3	5	A	P
8	Ritu is training a neural network, but she is not sure about her output layer. She is seeking help for the activation function. Differentiate softmax and sigmoid on 3 bases to help her realize which one would be suitable for her use case.	3	4	An	P
9	Why do we use a Pooling Layer in a CNN? Explain the significance of the RELU Activation function in Convolution Neural Network.	3	5	U	C
10	What is vanishing gradient problem? Mention one reason why it occurs and one solution to curb it.	3	4	U	F

Section – C

*Attempt **All** Questions* 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	**Height (H)** | **Gender(G)** | **Classification** 3 | F | Class 1 4 | F | Class 1 6 | M | Class 2 5 | M | Class 2 Note: For Class 1 probability <= 0.5 and Class 2 probability > 0.5 Consider person's Height dataset to predict the gender depending on Height. Apply Logistic regression to predict	5	2	A	P

	Gender of a person with Height = 2 and 8. Calculate it using line equation: Z = W*X + C Where W = -0.4 and C = 2.3				
12	Given the dimensions of the layers of a neural network, Derive y_hat(output) in terms of weights, biases, and the activations of respective layers as a forward propagation mechanism. Dimensions: Input = 5 Hidden Layer 1 = 5 Hidden Layer 2 = 4 Hidden Layer 3 = 3 Hidden Layer 4 = 2 Output Layer =1 where, Wi = Weight matrix of layer I, Bi = Bias matrix of layer I, Ai = Neuron matrix of hidden layer I, Activation Function = Sigmoid(σ) X = input Layer y_hat = σ (W5* σ (W4* σ (W3* σ (W2*(σ (W1*X + B1)+B2)+B3)+B4)+B5) **OR** What exactly does the term 'deep' in deep learning stand for? Whether true essence of deep learning lies in (i) Use of large number of layers, (ii) Automatic extraction of abstract features from raw data, or (iii) Both. Give your opinion with reasons.	5	4	An	P

CO1: Profound skills for creativity and innovation in automated organizational systems.
CO2: The enhanced ability to understand the creativity and innovation at workplace.
CO3: Well-constructed professional approach to understand the digital transformation and its ethical implications.
CO4: Improved Decision Making Ability for the successful leadership and innovation management skills

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End Term Examination, Odd Semester 2022-23
B. Tech CSE (H) II Year IV Semester
BELG0001, CREATIVITY AND INNOVATION

Time: 3 Hours **Maximum Marks: 45**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	There is seasonal product available with some company but they are lacking with space. They tried to relocate the product with available warehouse but they are not ready to give the service. Also they can't use off-site storage as the product is seasonal and shelf life of the product is very less. Based on the scenario given above, describe how the user was able to used creativity to solve the problem.	4	1	An	C
2	A digital transformation professional should be technically sound and good with people to tackle technical and people problems. What is more important, according to you, being technologically fluent or excellent at working with people?	4	2	R	F
3	Self-driving cars, collaborative production assistants, and socially-enabled domestic robots are operating in complex and open-ended environments with high levels of independence and self-determination. Explain the autonomous system operation involved in above	4	2	An	C
4	In an industry people are using machine to ease their operation. In another scenario a home maker using humanoid machine which is helping for daily household operation. Identify which from above has robotics and which has automation application, justify your answer with proper logic behind it.	4	4	A	P

5	List the industrial intelligent transportation systems and explain each in detail?	4	3	U	F

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Create a logic table for Line Follower Robot?	3	2	A	P
7	What is Arduino board? Explain the various components of Arduino?	3	3	R	F
8	Highlight the process involved in the implementation of any innovative product management.(Make your own assumption to decide the product)	3	3	An	C
9	Differentiate the term Robotics and Automation with reference to creativity and innovation.	3	4	R	F
10	Enlist at least two real life applications of Autonomous System.	3	4	U	C

Section – C

*Attempt **All** Questions* 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Supporting a creative work environment encourages innovation among your employees. Briefly suggest the idea to support creativity, compatibility of what have to offer and understanding of what they need to be creative?	5	3	A	P
12	Practice creativity with your team to see what drives innovation for you. You can sequence the process, inserting different steps or designing a different process that works better for a specific task. Justify above statement with real life problems defining dominant steps involved to solve the problem in an innovative way.	5	4	An	C

Course Name: Software Engineering

Course Outcome

- CO1: Understand the basic concepts of software engineering.
- CO2: Apply software processes to solve real world problems.
- CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
- CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
- CO5: Understand the basic concepts of OO analysis and design.
- CO6: Develop the test cases to validate the software.
- CO7: Understand the basic models of software Quality and maintenance.

Printed Pages: 4 **University Roll No.....**

End Term Examination, Even Semester 2022-23
B.Tech (CSE, CCV, AIML, CSF, IIOT, DA) II Year IV Semester
BCSC 0009 - Software Engineering

Time: 3 Hours **Maximum Marks: 50**

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Do the Parametric Comparison between Waterfall, Prototyping, Iterative Enhancement and Spiral software process model.	4	2	An	C
2	Suppose you are working on Requirement Analysis & Specification phase of a project. And you are in need to create a joint team of customers and developers just to understand the expectations and propose a set of requirements. Now identify the best requirement elicitation technique in this case & what guidelines will you follow.	4	2	An	C
3	Assume you are a software engineer working on a project to develop an e-commerce platform for a company. The platform will allow customers to browse and purchase products online, while also enabling the company to manage its inventory and track sales. Create a use case diagram that depicts the actors, use cases, and their relationships.	4	4	An	C
4	Consider a program for the determination of the nature of roots of a quadratic equation. Its input is a triple of positive integers (say a, b, c) and values	4	6	E	C

	may be from interval [0,100]. The program output may have one of the following words.[Not a quadratic equation; Real roots; Imaginary roots; Equal roots] Design the Boundary value test cases.				
5	Give Reason, "Maintenance process becoming more challenging than new development". Elaborate various categories of software maintenance process.	4	7	U	F

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Differentiate between A. Black Box Testing and Glass Testing B. Reverse Engineering and Re-engineering	3	6	U	C
7	Elucidate SEI-CMM Model of software quality. Explain how the CMM encourages continuous improvement of the software process by defining various levels.	3	6	U	C
8	An authorized User can withdraw or deposit the money using an ATM machine. Create sequence diagram by showing all the necessary steps of automatic dispensing of cash in ATM machine. Also discuss the purpose of creating sequence diagram.	3	4	C	P
9	A. How Cause Effect Graph Technique is different from Boundary Value Analysis. B. The development effort for a project is 600 PMs. The empirically determined constant (K) of Belady and Lehman model is 0.5. The complexity of code is quite high and is equal to 7. Calculate the total effort expended (M) if maintenance team has reasonable level of understanding of the project (d=0.7).	1+2	7	E	P

10	State the objectives of modular software design. What are the effects of module coupling and cohesion on a software design? What problems are likely to arise if two modules have high coupling and low cohesion?	3	5	U	C

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL

11	Draw the Control flow graph for the given program. Find out all independent paths and Design the test cases that will guarantee that all statements in the program have been tested. Also verify your answer by calculating Cyclomatic complexity.
	
	`#include <stdio.h>`
	`int main() {`
	
	`  //Here flag value 0 means prime number and`
	`  //1 means non-prime number`
	`int num, i, flag = 0;`
	`printf("Enter a number to check whether prime or not: ");`
	`  scanf("%d", &num);`
	
	`  // 0 and 1 are not the prime numbers so setting`
	`  // the flag to 1 if entered number is 0 or 1`
	`  if (num == 0 || num == 1)`
	`  flag = 1;`
	
	`  for (i = 2; i <= num / 2; ++i) {`
	
	`    // if num is divisible by i, then num is not prime`
	`    // change the flag value to 1`
	`    if (num % i == 0) {`
	`      flag = 1;`
	`      break;`
	`    }`
	`  }`
	
	`  // If flag is 0 it means the number is prime`
	`  if (flag == 0)`
	`    printf("Entered number %d is a prime number.", num);`
		5	6	E	P

	else printf("Entered number %d is not a prime number.", num); return 0; }				
12	Define Token Count. Consider a program (given in above Question 11 of Section C). List out the operators and operands available in program. Calculate the values of Halstead software science measures like η, N, V.	5	5	E	P
13	A. Discuss the purpose of Taute maintenance model and explain various phases of this model? B. Consider a software project has development effort of 500 PM. It is assumed that 10% code will be modified per year. Some of the cost multipliers are given as: 1.Required software Reliability (RELY):high= 1.15 2. Date base size (DATA) : high= 0.86 3. Analyst capability (ACAP) : high= 0.82 Other multipliers are nominal. Calculate the Annual Maintenance Effort (AME) using Boehm Maintenance model.	2+3	7	E	P

Course Name: OPERATING SYSTEM AND CONCEPTS

Course Outcome

CO1: Understand the classification of operating system environment.
CO2: Understand the basic of process management.
CO3: Apply the concept of CPU process scheduling for the given scenarios.
CO4: Illustrate the process synchronization and concurrency process in operating system.
CO5: Analyze the occurrence of deadlock in operating system.
CO6: Describe and analyze the memory management and its allocation policies.
CO7: Understand the concepts of disk scheduling.

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End Term Examination, Even Semester 2022-23
B.Tech. (Hons) CSE, 2nd, VI Semester
BCSC 0055: OPERATING SYSTEM AND CONCEPTS

Time: 3 Hours **Maximum Marks: 45**

Instruction for students:

- Attempt all questions.
- It is advisable to attempt all questions of section all together.
- Draw neat and clean diagram, wherever required.

Section – A

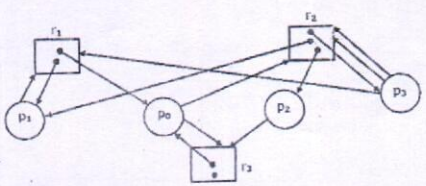
Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Illustrate the differences between symmetric and asymmetric multiprocessing. Derive the advantages and disadvantages of multiprocessor systems.	4	1	U	C
2	Show how counting semaphores can be implemented using only binary semaphores and ordinary machine instructions. Or Demonstrate the use of IPC in terms of withdrawing money from ATM. Which fundamental model of IPC is used in this scenario, illustrate through draw the use case for scenario *money withdrawal from ATM.*	4	3	A	C
3	Students working at individual PCs in a computer laboratory send their files to be printed by a server which spools the files on its hard disk. Under what conditions may a deadlock occur if the disk space for the print spool is limited? How may the deadlock be avoided?	4	5	An	M

4	Suppose we have a demand paged memory. The page table is held in registers. it takes 8ms to service a page fault if an empty page is available or the replaced page is not modified, and 20ms if the replaced page is modified. memory access time is 100ns. Assume that the page to be replaced is modified 70% of the time. what is the maximum acceptable page fault rate for an effective access time of no more than 200ns?	4	6	A	P
5	Enumerate the term locality of reference and elaborate on its usefulness in presenting thrashing. Elucidate how does the system recover from thrashing?	4	6	R	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the resource allocation graph given in the figure.  a) Find if the system is in a deadlock state. b) Otherwise, find a safe sequence.	3	5	A	P
7	Compare the memory organization schemes of contiguous memory allocation, pure segmentation, and pure paging with respect to the following issues: a) External Fragmentation b) Internal Fragmentation c) Ability to share code across processes	3	6	U	C
8	Can you think of any situations where supporting virtual memory would be a bad idea, and what would be gained by not having to support virtual memory? Clarify this statement.	3	6	An	C
9	A process references 5 pages A, B , C, D, E in the following order A, B, C, D, A, E, B, C, E, D Assuming that the replacement algorithm is LRU	3	6	A	P

	and FIFO, find out the number of page faults during the sequence of references, starting with an empty main memory With 3 frames.				
10	Suppose that the head of moving head disk with 200 tracks numbered 0 to 199 is currently serving the request at track 143 and has just finished a request at track 125. If the queue request is kept in FIFO order, 86, 147, 91, 177, 94, 150, 102, 175, 130. What is the total head movement to satisfy these requests for i) FCFS II) SSTF disk scheduling algorithm.	3	7	A	P

Section – C

*Attempt **All** Questions* 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Five batch jobs A through E arrives at a computer center at almost the same time. They have estimated running times of 10, 6, 2, 4, and 8 minutes. Their priorities are 3, 5, 2, 1, and 4, respectively, with 5 being the highest priority. For each of the following scheduling algorithms, determine the mean process turn-around time. Ignore process switching overhead. a) Round Robin (Time slice 2 minutes) b) Shortest Job First c) Priority Scheduling d) First-come, first-served.	5	3	A	P
12	What would happen if the bitmap or free list containing the information about free space on disk blocks was completely lost due to a crash? Is there any way to recover from this disaster, or is it bye-bye disk? Deliberate your answer for Windows NTFS system. Or Contrast the performance of three techniques for allocating disk blocks for both sequential and random file access.	5	7	U	C

Course Name: Software Engineering Methodologies BCSC0057

Course Outcome

CO1: Understand the basic concepts of software engineering.
CO2: Apply software processes to solve real world problems.
CO3: Estimate the cost, effort and schedule of software using COCOMO Model.
CO4: Analyze the software design techniques (structure chart, SDM, sequence diagram).
CO5: Understand the basic concepts of OO analysis and design.
CO6: Develop the test cases to validate the software.
CO7: Understand the basic models of software Quality and maintenance.

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End Term Examination, Even Semester 2022-23
B. Tech. (Hons.) CS, Year: II, Semester: IV
Software Engineering Methodologies BCSC0057

Time: 3 Hours **Maximum Marks: 45**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Compare the Walston-Felix model with the SEL model for a software development which is expected to involve 20 person-years of effort by calculating the following: a) The number of lines of source code that can be produced. b) The duration of the development. c) The productivity in LOC/PY d) Average manning	4	3	E	M
2	Describe the characteristics of software contrasting it with the characteristics of hardware. Also Explain Agile Framework with suitable Diagram and advantages. **Or** How to ensure optimal number of modules during modularity. Enlist various types of coupling and cohesion and explain any three types of coupling and cohesion schemes.	4	1	R	F
3	Draw an activity diagram and sequence diagram for the following specification. A customer sends the order for request; the order request system confirms the receipt of the order; the customer makes the payment. If the payment is received, then order is	4	4	An	D

	confirmed and product is shipped to the customer otherwise order is closed.				
4	Discuss software maintenance process. The development effort for a project is 800 PMs. The empirically determined constant (K) of Belady and Lehman model is 0.7. The complexity of code is quite high and is equal to 9. Calculate the total effort expended (M) if a) Maintenance team has reasonable level of understanding of project (d=0.7). b) Maintenance team has poor understanding of project (d=0.1).	4	7	E	M
5	Differentiate between Mc Call Model and Boehm Model for software quality. Enlist various key process areas of SEI-CMM at various maturity levels.	4	7	A	P

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Draw the framework of Automation testing. Why Automation Testing is needed? Explain Automated Testing process of Selenium with the help of examples (By considering different test cases).	3	6	C	M
7	Explain Test Driven Development approach of coding with diagram. How it is different from incremental coding process.	3	7	U	F

8	Compute vocabulary, program length, volume of the given code:
	#include <stdio.h>
	int main()
	{
	int i, n, t1 = 0, t2 = 1, nextTerm, nex;
	printf("Enter the number of terms");
	scanf("%d", &n);
	printf("Fibonacci Series ");
	for (i = 1; i <= n; ++i)
	{
	printf("%d ", t1);
	nex= t2;
	t2 = nextTerm;
	}
	return 0;
		3	6	E	F

	}				
9	Explain Equivalence Class Testing Method. Consider the following scenarios and write the test cases according to Equivalence Class Testing Method for each scenarios. Assume, we have to test a field which accepts Age 18 – 56 for man and Age 20-52 for women. Assume, we have to test a filed which accepts a Mobile Number of ten digits. Suppose you have very important tool at office, accepts valid User Name and Password field to work on that tool, and accepts minimum 8 characters and maximum 12 characters. Valid range 8-12, Invalid range 7 or less than 7 and Invalid range 13 or more than 13.	3	5	E	F
10	Consider a program for determining the Previous date. Its input is a triple of day, month and year with the values in the range $1 \leq \text{month} \leq 12$; $1 \leq \text{day} \leq 31$; $1900 \leq \text{year} \leq 2025$ The possible outputs would be Previous date or invalid input date. Design the boundary value test cases.	3	6	C	D

Section – C

*Attempt **All** Questions* 5 X 2 = 10 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	For the following situation, draw a use case diagram, context-level diagram and a level-1 data flow diagram. Kellogg State Bank provides car and home loans to its banking customers. Initially, a potential loan customer meets with a Kellogg loan officer, requests a loan for a certain amount and time frame, and completes a loan application. Next, the loan officer determines the customer's credit standing, the type of loan required, and available interest rates. While the loan officer can authorize car loans for credit worthy customers, a loan committee must approve all home loans.	5	2	C	D
12	Define Cause effect graphing technique. Represent the following using a decision table. There are three types of tax-payers---male, female and senior citizen. A male tax-payer pays 10% tax for income of more than Rs 1,60,000 and less than Rs. 3,00,000. A female pays	5	6	R	F

10% tax, if her income is more than Rs. 1,80,000 and less than Rs. 3,00,000, and a senior citizen pays 10% tax, if his income is more than Rs. 2,10,000 and less than 3,00,000. If a tax -payer income is more than 3,00,000 and less than 6,00,000, he has to pay 10% tax up to 3,00,000 + 20% tax on the income which is more than 3,00,000. If the income is more than 5,00,000 and less than Rs. 10,00,000, then he has to pay tax up to income Rs. 5,00,000 + 30% tax on the income which is more than Rs. 5,00,000. If the income is more than Rs. 10,00,000, then a tax-payer has to pay tax on the income up to Rs. 10,00,000 + 40% tax on the income higher than1 0,00,000.

Or

Consider the program below-draw the flow graph and convert it to DD path graph. Estimate cyclometric complexity and independent path.

```
#include<stdio.h>
#include<conio.h>
1.void main()
2.{
3. float amount,dis,pur;
4. clrscr();
5. printf("enter the purchasing amount ");
6. scanf("%f",&pur);
7. if(pur>1000)
8. {
9. dis=pur*0.1;
10. }
11. else
12. {
13. dis=pur*0.05;
14. }
15. amount=pur-dis;
16. printf("purchase amount =%f\n",pur);
17. printf("discount =%f\n",dis);
18. printf("total amount =%f\n",amount);
19. getch();
20. }
```